

Weather and Sales Data Analysis using Hadoop MapReduce

Objective

The objective of this project is to implement Hadoop Streaming with Python-based Mapper and Reducer programs to process large datasets efficiently. The project demonstrates distributed data processing by performing:

- Weather data analysis to identify the hottest and coldest days.
- Sales data analysis to determine the top-selling products and the best-performing products in each city.

Project Description

This project uses the Hadoop MapReduce framework to analyze two different datasets.

- Exercise 1: A weather dataset containing date, city, maximum temperature, and minimum temperature is processed using Mapper and Reducer programs. The system identifies the hottest and coldest days from the dataset.
- Exercise 2: A sales dataset containing product sales information is analyzed to calculate total sales for each product, identify the top five products, and determine the highest-selling product for each city.

The datasets are uploaded to HDFS, processed using Hadoop Streaming, and the final results are stored and displayed from Hadoop output directories. Python is used to implement the Mapper and Reducer logic.

Code:

Exercise 1: Weather Data Analysis (Temperature Analysis) Objective

Analyze temperature data to find the hottest and coldest days using MapReduce. Dataset

(Create This)

```
bash
```

```
#Generate weather data for 1 year (365 days) cat
```

```
> weather_data.txt <<'EOF'
```

```
2024-01-01,Delhi,12,5
```

```
2024-01-02,Delhi,14,6
```

```
2024-01-03,Delhi,10,3
```

```
2024-01-04,Mumbai,28,20
```

```
2024-01-05,Mumbai,30,22
```

```

2024-01-06,Bangalore,25,18
2024-01-07,Bangalore,26,19
2024-01-08,Chennai,32,25
2024-01-09,Chennai,33,26
EOF
#Makeit larger (copy100 times)
foriin{1..100};docatweather_data.txt;done>weather_large.txt Data
Format: Date,City,MaxTemp,MinTemp
PartA:Mapper-ExtractMaxTemperature bash
cat>mapper_max_temp.py<<'EOF'
#!/usr/bin/env python3
import sys

forlinein sys.stdin:
    line=line.strip() if
    not line:
        continue
    parts=line.split(',') if
    len(parts) != 4:
        continue
    try:
        date=parts[0]
        city=parts[1]
        max_temp=float(parts[2])
        min_temp = float(parts[3])
        # Emit: date \t max_temp
        print(f"{date}\t{max_temp}")
    except ValueError:
        continue
EOF
chmod +x mapper_max_temp.py
PartB:Reducer-FindHottestDay
bash
cat>reducer_max_temp.py<<'EOF'
#!/usr/bin/env python3
import sys

max_temp=-float('inf')
hottest_date = None
forlineinsys.stdin:
    line = line.strip()
    ifnotline:
        continue

```

```

date,temp=line.split('\t') temp
= float(temp)
if temp > max_temp:
    max_temp = temp
    hottest_date=date
if hottest_date is not None:
    print(f"HottestDay:{hottest_date} with {max_temp}°C") EOF
chmod+x reducer_max_temp.py

```

PartC:RuntheJob

bash

#Upload to HDFS

hdfs dfs -mkdir -p /user/\$USER/weather/input

hdfs dfs -put weather_large.txt /user/\$USER/weather/input/

Run MapReduce

hadoop jar \$HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \

-input /user/\$USER/weather/input/weather_large.txt \

-output /user/\$USER/weather/output \

-mapper "python3 mapper_max_temp.py" \

-reducer "python3 reducer_max_temp.py" \

-file mapper_max_temp.py \

-file reducer_max_temp.py

View result

hdfs dfs -cat /user/\$USER/weather/output/part-00000 Part

D: Coldest Day Analysis

bash

Create a reducer for coldest day

cat > reducer_min_temp.py <<'EOF'

#!/usr/bin/env python3

import sys

min_temp=float('inf')

coldest_date = None

for line in sys.stdin:

line = line.strip()

if not line:

continue

date,temp= line.split('\t')

temp =

float(temp) if temp <

min_temp:

min_temp = temp

coldest_date = date

```

ifcoldest_dateisnotNone:
print(f"ColdestDay:{coldest_date}with{min_temp}°C") EOF
chmod+xreducer_min_temp.py #
Run
hadoopjar$HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar\
-input/user/$USER/weather/input/weather_large.txt\
-output/user/$USER/weather/output_min\
-mapper"python3mapper_max_temp.py"\
-reducer"python3reducer_min_temp.py"\
-filemapper_max_temp.py \
-filereducer_min_temp.py
hdfsdfs-cat/user/$USER/weather/output_min/part-00000 Expected
Output
text
HottestDay:2024-06-15with45.5°C
ColdestDay:2024-01-15with-2.0°C

```

Exercise2:SalesAnalysis-TopProducts

Objective

Findthetop 5productsbytotal sales.

Dataset

bash

#Generatesalesdatawithproducts

cat > sales_products.txt

<<'EOF'2024-01-

01,Laptop,45000,Delhi 2024-01-

01,Mouse,1500,Mumbai

2024-01-02,Keyboard,3500,Bangalore

2024-01-02,Laptop,45000,Chennai

2024-01-03,Mouse,1500,Delhi

2024-01-03,Monitor,25000,Mumbai

2024-01-04,Laptop,45000,Bangalore

2024-01-04,Keyboard,3500,Delhi

2024-01-05,Monitor,25000,Chennai

2024-01-05,Mouse,1500,Mumbai

EOF

#Makeit larger

foriin{1..50};docatsales_products.txt;done>sales_products_large.txt Data

Format: Date,Product,Price,City

PartA:Mapper-ExtractProductSales bash

cat>mapper_top_products.py<<'EOF'

```
#!/usr/bin/envpython3
import sys

forlinein sys.stdin:
line=line.strip() if
not line:
continue
parts=line.split(',') if
len(parts) != 4:
continue
try:
date = parts[0]
product = parts[1]
price=float(parts[2])
city = parts[3]
# Emit: product \t price
print(f"{product}\t{price}")
except ValueError:
continue
EOF
chmod +x
mapper_top_products.pyPartB:Reducer
-SumSalesbyProduct bash
cat>reducer_top_products.py<<'EOF'
#!/usr/bin/env python3
```

```
import sys
current_product=None
total_sales = 0
product_sales = []
forlinein sys.stdin:
line=line.strip() if
not line:
continue
product,price=line.split("\t")
price = float(price)
ifproduct== current_product:
total_sales+=price
else:
if current_product is not None:
product_sales.append((total_sales,current_product))
current_product = product
total_sales=price #
Last product
```

```

if current_product is not None:
    product_sales.append((total_sales,current_product))
# Sort by sales (descending) and get top 5
product_sales.sort(reverse=True)

print("TOP5PRODUCTSBYSALES:")
print("="* 40)
for i,(sales,product)inenumerate(product_sales[:5],1): print(f"{i}.
{product}: ₹{sales:,.2f}")
EOF
chmod+xreducer_top_products.py

```

Part C: Run the Job

```

bash
#Upload to HDFS
hdfsdfs-mkdir-p/user/$USER/sales_products/input
hdfsdfs-putsales_products_large.txt/user/$USER/sales_products/input/ #
Run MapReduce
hadoopjar$HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar\
-input/user/$USER/sales_products/input/sales_products_large.txt\
-output/user/$USER/sales_products/output\
-mapper"python3mapper_top_products.py"\
-reducer"python3reducer_top_products.py"\
-filemapper_top_products.py \
-filereducer_top_products.py
# View results
hdfsdfs-cat/user/$USER/sales_products/output/part-00000

```

Part D: Advanced - City-wise Top Products

```

bash
cat>mapper_city_products.py<<'EOF'
#!/usr/bin/env python3
import sys
for line in sys.stdin:
    line = line.strip()
    if not line:
        continue
    parts=line.split(',')
    if len(parts) != 4:
        continue
    try:
        date = parts[0]
        product = parts[1]
        price=float(parts[2])
        city = parts[3]

```

```
# Emit: city_product \t price
print(f"{city}_{product}\t{price}")
except ValueError:
    continue
EOF
chmod+x mapper_city_products.py
```

```
cat>reducer_city_products.py<<'EOF'
#!/usr/bin/env python3
import sys
current_key=None
total_sales = 0
results = {}
for line in sys.stdin:
    line=line.strip()
    if not line:
        continue
    key,price=line.split('\t')
    price = float(price)
    if key== current_key:
        total_sales+=price
    else:
        if current_key is not None:
            city,product=current_key.split('_')
            if city not in results:
                results[city] = []
            results[city].append((total_sales,product))
            current_key = key
            total_sales=price #
            # Last key
        if current_key is not None:
            city,product=current_key.split('_')
            if city not in results:
                results[city] = []
            results[city].append((total_sales,product))
        # Print top products per city
        print("TOPPRODUCTPERCITY:")
        print("="* 40)
        for city,products in results.items():
            products.sort(reverse=True)
            top_product = products[0]
            print(f"{city}:{top_product[1]}({sum(top_product[0]:,.2f)})")
        EOF
```

```

chmod +x reducer_city_products.py
CompleteScripttoRunBothExercises bash
#!/bin/bash
#Runbotheexerciseseinsequence
echo"=====
echo "EXERCISE 1: WEATHER DATA ANALYSIS"
echo"=====
#Uploadweatherdata
hdfsdfs-mkdir-p/user/$USER/weather/input

hdfsdfs-putweather_large.txt/user/$USER/weather/input/ #
Run max temperature
hadoopjar$HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar\
-input/user/$USER/weather/input/weather_large.txt\
-output/user/$USER/weather/output_max\
-mapper"python3mapper_max_temp.py"\
-reducer"python3reducer_max_temp.py"\
-filemapper_max_temp.py \
-file reducer_max_temp.py
echo-e"\n---MAXTEMPERATURE---"
hdfsdfs-cat/user/$USER/weather/output_max/part-00000
echo-e"\n=====
echo "EXERCISE 2: TOP PRODUCTS"
echo"=====
#Uploadsalesdata
hdfsdfs-mkdir-p/user/$USER/sales_products/input
hdfsdfs-putsales_products_large.txt/user/$USER/sales_products/input/ #
Run top products
hadoopjar$HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar\
-input/user/$USER/sales_products/input/sales_products_large.txt\
-output/user/$USER/sales_products/output\
-mapper"python3mapper_top_products.py"\
-reducer"python3reducer_top_products.py"\

-filemapper_top_products.py \
-filereducer_top_products.py
echo-e"\n---TOPPRODUCTS ---"
hdfsdfs-cat/user/$USER/sales_products/output/part-00000
echo -e "\n Both exercises completed!"

```

Screenshot

```
sky@MS97377: ~/Exercise6
sky@MS97377:~$ jps
719 Jps
sky@MS97377:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [MS97377]
sky@MS97377:~$ start-yarn.sh
start.yarn.sh: command not found
sky@MS97377:~$ start-yarn.sh
Starting resourceManager
Starting nodeManagers
sky@MS97377:~$ jps
883 NameNode
1161 SecondaryNameNode
1547 NodeManager
1436 ResourceManager
1004 DataNode
1933 Jps
sky@MS97377:~$ mkdir Exercise6
sky@MS97377:~$ cd Exercise6
sky@MS97377:~/Exercise6$ cat > weather_data.txt << 'EOF'
> 2024-01-01,Delhi,12,5
2024-01-02> 2024-01-02,Delhi,14,6
> 2024-01-03,Delhi,10,3
> 2024-01-04,Mumbai,28,20
2024-01-05,> 2024-01-05,Mumbai,30,22
> 2024-01-06,Bangalore,25,18
> 2024-01-07,Bangalore,26,19
> 2024-01-08,Chennai,32,25
> 2024-01-09,Chennai,33,26
> EOF
sky@MS97377:~/Exercise6$ for i in {1..100}; do
> cat weather_data.txt
> done > weather_large.txt
sky@MS97377:~/Exercise6$ nano mapper_max_temp.py
sky@MS97377:~/Exercise6$ chmod +x mapper_max_temp.py
sky@MS97377:~/Exercise6$ nano reducer_max_temp.py
sky@MS97377:~/Exercise6$ nano reducer_max_temp.py
sky@MS97377:~/Exercise6$ chmod +x reducer_max_temp.py
sky@MS97377:~/Exercise6$ hdfs dfs -mkdir -p /user/$USER/weather/input
sky@MS97377:~/Exercise6$ hdfs dfs -put -f weather_large.txt /user/$USER/weather/input/
sky@MS97377:~/Exercise6$ hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
> -input /user/$USER/weather/input/weather_large.txt \
> -output /user/$USER/weather/output \
Map-Reduce Framework
CPU time spent (ms)=0
Physical memory (bytes) snapshot=0
Virtual memory (bytes) snapshot=0
2026-07-25 22:52:50,868 ERROR streaming.StreamJob: Job not successful!
Streaming Command Failed!
sky@MS97377:~/Exercise6$ cat weather_large.txt | python3 mapper_max_temp.py
File "/home/sky/Exercise6/mapper_max_temp.py", line 5
    continue
    ^^^^^^^^^
SyntaxError: 'continue' not properly in loop
sky@MS97377:~/Exercise6$ python3 mapper_max_temp.py < weather_large.txt
File "/home/sky/Exercise6/mapper_max_temp.py", line 5
    continue
    ^^^^^^^^^
SyntaxError: 'continue' not properly in loop
sky@MS97377:~/Exercise6$ nano mapper_max_temp.py
sky@MS97377:~/Exercise6$ chmod +x mapper_max_temp.py
sky@MS97377:~/Exercise6$ python3 mapper_max_temp.py < weather_large.txt
2024-01-01 12.0
2024-01-02 14.0
2024-01-03 10.0
2024-01-04 28.0
2024-01-05 30.0
2024-01-06 25.0
2024-01-07 26.0
2024-01-08 32.0
2024-01-09 33.0
2024-01-01 12.0
2024-01-02 14.0
2024-01-03 10.0
2024-01-04 28.0
2024-01-05 30.0
2024-01-06 25.0
2024-01-07 26.0
2024-01-08 32.0
2024-01-09 33.0
2024-01-01 12.0
2024-01-02 14.0
2024-01-03 10.0
2024-01-04 28.0
2024-01-05 30.0
2024-01-06 25.0
2024-01-07 26.0
```

```

sky@MS97377: ~/Exercise6
2024-01-02 14.0
2024-01-03 10.0
2024-01-04 28.0
2024-01-05 30.0
2024-01-06 25.0
2024-01-07 26.0
2024-01-08 32.0
2024-01-09 33.0
2024-01-01 12.0
2024-01-02 14.0
2024-01-03 10.0
2024-01-04 28.0
2024-01-05 30.0
2024-01-06 25.0
2024-01-07 26.0
2024-01-08 32.0
2024-01-09 33.0
2024-01-01 12.0
2024-01-02 14.0
2024-01-03 10.0
2024-01-04 28.0
2024-01-05 30.0
2024-01-06 25.0
2024-01-07 26.0
2024-01-08 32.0
2024-01-09 33.0
sky@MS97377:~/Exercise6$ hdfs dfs -rm -r /user/$USER/weather/output
Deleted /user/sky/weather/output
sky@MS97377:~/Exercise6$ hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
> -input /user/$USER/weather/input/weather_large.txt \
> -output /user/$USER/weather/output \
> -mapper "python3 mapper_max_temp.py" \
> -reducer "python3 reducer_max_temp.py" \
> -file mapper_max_temp.py \
> -file reducer_max_temp.py
2026-07-25 22:56:53,830 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
packageJobJar: [mapper_max_temp.py, reducer_max_temp.py, /tmp/hadoop-unjar5668236442819812528/] [] /tmp/streamjob8678324389109777462.jar tmpDir=null
2026-07-25 22:56:55,479 INFO client.DefaultHARMAFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-07-25 22:56:55,821 INFO client.DefaultHARMAFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-07-25 22:56:56,157 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/sky/.staging/job_1784989774809_0002
2026-07-25 22:56:57,759 INFO mapred.FileInputFormat: Total input files to process : 1
2026-07-25 22:56:58,386 INFO mapreduce.JobSubmitter: number of splits:2
2026-07-25 22:56:59,115 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1784989774809_0002
2026-07-25 22:56:59,115 INFO mapreduce.JobSubmitter: Executing with tokens: []
sky@MS97377:~/Exercise6
2026-07-25 22:57:47,088 ERROR streaming.StreamJob: Job not successful!
Streaming Command Failed!
sky@MS97377:~/Exercise6$ cat weather_large.txt | python3 mapper_max_temp.py | sort | python3 reducer_max_temp.py
File "/home/sky/Exercise6/reducer_max_temp.py", line 3
    max_temp = -float('inf')
IndentationError: unexpected indent
sky@MS97377:~/Exercise6$ python3 reducer_max_temp.py
File "/home/sky/Exercise6/reducer_max_temp.py", line 3
    max_temp = -float('inf')
IndentationError: unexpected indent
sky@MS97377:~/Exercise6$ cat reducer_max_temp.py
#!/usr/bin/env python3
import sys
max_temp = -float('inf')
hottest_date = None
for line in sys.stdin:
    line = line.strip()
    if not line:
        continue
    date, temp = line.split('\t')
    temp = float(temp)
    if temp > max_temp:
        max_temp = temp
        hottest_date = date
if hottest_date is not None:
    print(f'Hottest Day: {hottest_date} with {max_temp}°C')
EOF
sky@MS97377:~/Exercise6$ nano reducer_max_temp.py
sky@MS97377:~/Exercise6$ chmod +x reducer_max_temp.py
sky@MS97377:~/Exercise6$ cat weather_large.txt | python3 mapper_max_temp.py | sort | python3 reducer_max_temp.py
Hottest Day: 2024-01-09 with 33.0°C
sky@MS97377:~/Exercise6$ hdfs dfs -rm -r /user/$USER/weather/output
Deleted /user/sky/weather/output
sky@MS97377:~/Exercise6$ hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
> -input /user/$USER/weather/input/weather_large.txt \
> -reducer -output /user/$USER/weather/output \
> -mapper "python3 mapper_max_temp.py" \
> -reducer "python3 reducer_max_temp.py" \
> -file mapper_max_temp.py \
> -file reducer_max_temp.py
2026-07-25 23:02:07,577 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
packageJobJar: [mapper_max_temp.py, reducer_max_temp.py, /tmp/hadoop-unjar4758276041923459360/] [] /tmp/streamjob2934787196920328744.jar tmpDir=null
2026-07-25 23:02:09,126 INFO client.DefaultHARMAFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-07-25 23:02:09,451 INFO client.DefaultHARMAFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-07-25 23:02:09,761 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/sky/.staging/job_1784989774809_0003
2026-07-25 23:02:10,818 INFO mapred.FileInputFormat: Total input files to process : 1
2026-07-25 23:02:10,976 INFO mapreduce.JobSubmitter: number of splits:2
2026-07-25 23:02:11,175 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1784989774809_0003
2026-07-25 23:02:11,176 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-07-25 23:02:11,457 INFO conf.Configuration: resource-types.xml not found
2026-07-25 23:02:11,458 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-07-25 23:02:11,594 INFO impl.YarnClientImpl: Submitted application application_1784989774809_0003
2026-07-25 23:02:11,659 INFO mapreduce.Job: The url to track the job: http://MS97377.localdomain:8088/proxy/application_1784989774809_0003/
2026-07-25 23:02:11,663 INFO mapreduce.Job: Running job: job_1784989774809_0003
2026-07-25 23:02:23,063 INFO mapreduce.Job: Job job_1784989774809_0003 running in uber mode : false
2026-07-25 23:02:23,068 INFO mapreduce.Job: map 0% reduce 0%
2026-07-25 23:02:31,307 INFO mapreduce.Job: map 50% reduce 0%
2026-07-25 23:02:32,329 INFO mapreduce.Job: map 100% reduce 0%
2026-07-25 23:02:39,487 INFO mapreduce.Job: map 100% reduce 100%
2026-07-25 23:02:40,463 INFO mapreduce.Job: Job job_1784989774809_0003 completed successfully
2026-07-25 23:02:40,669 INFO mapreduce.Job: Counters: 54
File System Counters
FILE: Number of bytes read=16206
FILE: Number of bytes written=870623
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=26124
HDFS: Number of bytes written=38
HDFS: Number of read operations=11
HDFS: Number of large read operations=0

```

```

sky@MS97377: ~/Exercise6
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=25896
File Output Format Counters
  Bytes Written=38
2026-07-25 23:02:40,670 INFO streaming.StreamJob: Output directory: /user/sky/weather/output
sky@MS97377:~/Exercise6$ hdfs dfs -cat /user/$USER/weather/output/part-00000
Hottest Day: 2024-01-09 with 33.0°C
sky@MS97377:~/Exercise6$ cat > sales_products.txt << 'EOF'
> 2024-01-01,Laptop,45000,Delhi
> 2024-01-01,Mouse,1500,Mumbai
> 2024-01-02,Keyboard,3500,Bangalore
> 2024-01-02,Laptop,45000,Chennai
> 2024-01-03,Mouse,1500,Delhi
> 2024-01-03,Monitor,25000,Mumbai
> 2024-01-04,Laptop,45000,Bangalore
> 2024-01-04,Keyboard,3500,Delhi
> 2024-01-05,Monitor,25000,Chennai
> 2024-01-05,Mouse,1500,Mumbai
EOF
sky@MS97377:~/Exercise6$ for i in {1..50}; do
> cat sales_products.txt
> done > sales_products_large.txt
sky@MS97377:~/Exercise6$ nano mapper_top_products.py
sky@MS97377:~/Exercise6$ chmod +x mapper_top_products.py
sky@MS97377:~/Exercise6$ python3 mapper_top_products.py < sales_products_large.txt
Laptop 45000.0
Mouse 1500.0
Keyboard 3500.0
Laptop 45000.0
Mouse 1500.0
Monitor 25000.0
Laptop 45000.0
Keyboard 3500.0
Monitor 25000.0
Mouse 1500.0
Laptop 45000.0
Mouse 1500.0
Keyboard 3500.0
Laptop 45000.0
Mouse 1500.0
Monitor 25000.0
sky@MS97377:~/Exercise6$ nano reducer_top_products.py
sky@MS97377:~/Exercise6$ chmod +x reducer_top_products.py
sky@MS97377:~/Exercise6$ cat sales_products_large.txt | python3 mapper_top_products.py | sort | python3 reducer_top_products.py
TOP 5 PRODUCTS BY SALES
=====
Laptop ₹6750000.00
Monitor ₹2500000.00
Keyboard ₹350000.00
Mouse ₹225000.00
sky@MS97377:~/Exercise6$ hdfs dfs -mkdir -p /user/$USER/sales_products/input
sky@MS97377:~/Exercise6$ hdfs dfs -put -f sales_products_large.txt /user/$USER/sales_products/input/
sky@MS97377:~/Exercise6$ hdfs dfs -ls /user/$USER/sales_products/input
Found 1 items
-rw-r--r-- 1 sky supergroup 15650 2026-07-25 23:06 /user/sky/sales_products/input/sales_products_large.txt
sky@MS97377:~/Exercise6$ hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
put /use> -input /user/$USER/sales_products/input/sales_products_large.txt \
> -output /user/$USER/sales_products/output \
> -mapper "python3 mapper_top_products.py" \
> -reducer "python3 reducer_top_products.py" \
> -file mapper_top_products.py \
reducer> -file reducer_top_products.py
2026-07-25 23:07:11,153 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
packageJobOptions: [mapper_top_products.py, reducer_top_products.py, /tmp/hadoop-unjam670535890034493576/[] /tmp/streamjob8249243123265954953.jar tmpDir=null
2026-07-25 23:07:12,747 INFO client.DefaultHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-07-25 23:07:13,072 INFO client.DefaultHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-07-25 23:07:13,390 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/sky/.staging/job_1784989774809_0004
2026-07-25 23:07:14,567 INFO mapred.FileInputFormat: Total input files to process : 1
2026-07-25 23:07:14,698 INFO mapreduce.JobSubmitter: number of splits:2
sky@MS97377:~/Exercise6
Failed Shuffles=0
Merged Map outputs=2
GC time elapsed (ms)=1214
CPU time spent (ms)=5100
Physical memory (bytes) snapshot=1382989824
Virtual memory (bytes) snapshot=7640125440
Total committed heap usage (bytes)=1196425216
Peak Map Physical memory (bytes)=564084736
Peak Map Virtual memory (bytes)=2546372608
Peak Reduce Physical memory (bytes)=264454144
Peak Reduce Virtual memory (bytes)=2551091200
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=19746
File Output Format Counters
  Bytes Written=141
2026-07-25 23:07:42,696 INFO streaming.StreamJob: Output directory: /user/sky/sales_products/output
sky@MS97377:~/Exercise6$ hdfs dfs -cat /user/$USER/sales_products/output/part-00000TOP 5 PRODUCTS BY SALES
=====
Laptop ₹6750000.00
Monitor ₹2500000.00
Keyboard ₹350000.00
Mouse ₹225000.00
sky@MS97377:~/Exercise6$

```

Conclusion

This project demonstrates the practical implementation of the Hadoop MapReduce framework for processing large-scale datasets. By using Python-based Mapper and Reducer programs, weather data was analysed to determine the hottest and coldest days, while sales data was processed to identify the top-selling products and city-wise product performance. The project highlights the efficiency of distributed computing in handling big data and shows how Hadoop Streaming can be integrated with Python to perform scalable data analysis. Overall, the exercises provide a strong understanding of MapReduce programming, HDFS operations, and real-world big data analytics using Hadoop.